

Non-intrusive R_{ds-on} Measurement for SiC using Gate Driver Measurement Units

Aim of the Study

The aim of this study is to develop a non-intrusive method for estimating the on-state resistance (R_{ds-on}) of Silicon Carbide (SiC) MOSFETs using additional measurement units integrated within the gate driver. The objective is to achieve accurate R_{ds-on} measurement without direct current sensing, enabling real-time condition monitoring of SiC MOSFETs.

Motivation

- R_{ds-on} is an important health indicator for SiC MOSFETs, as it correlates with gate-oxide degradation and junction temperature changes.
- Traditional methods rely on direct current measurement, which can be intrusive and impractical for real-time applications.
- A non-intrusive approach would enhance the reliability and efficiency of power converters using SiC technology.

Objectives

- Develop a theoretical model for non-intrusive R_{ds-on} measurement using additional gate driver units.
- Design a circuit incorporating high-frequency current injection and differential voltage sensing.
- Validate the approach through simulations.
- Optional: Validate the approach through experiments.
- Optional: Analyze the correlation between R_{ds-on} and device health indicators such as gate-oxide degradation.

Methodology

- Literature review of existing R_{ds-on} measurement techniques and health monitoring methods for SiC MOSFETs.
- Theoretical development of a non-intrusive measurement method using impedance measurement principles.
- Circuit design and simulation to optimize accuracy and minimize interference with normal converter operation.
- Optional: Experimental validation using double pulse test (DPT) setup.

Expected Contributions

- A non-intrusive R_{ds-on} measurement technique for SiC MOSFETs without direct current sensing.
- Real-time health monitoring capability for power converters using SiC technology.
- Optional: Insights into the relationship between R_{ds-on} and device aging indicators.

Initial Literature

- F. Karakaya, A. Maheshwari, A. Banerjee and J. S. Donnal, "An Approach for Online Estimation of On-State Resistance in SiC MOSFETs Without Current Measurement," in IEEE Transactions on Power Electronics, vol. 38, no. 9, pp. 11463-11473, Sept. 2023, doi: 10.1109/TPEL.2023.3282585.
- J. Huang et al., "Detection of on-State Impedance Based on High-Frequency Current Injection Method," 2024 3rd International Symposium on Semiconductor and Electronic Technology (ISSET), Xi'an, China, 2024, pp. 405-409, doi: 10.1109/ISSET62871.2024.10779790.